

Evaluating Age Estimation Robustness Under Realistic Facial Occlusions

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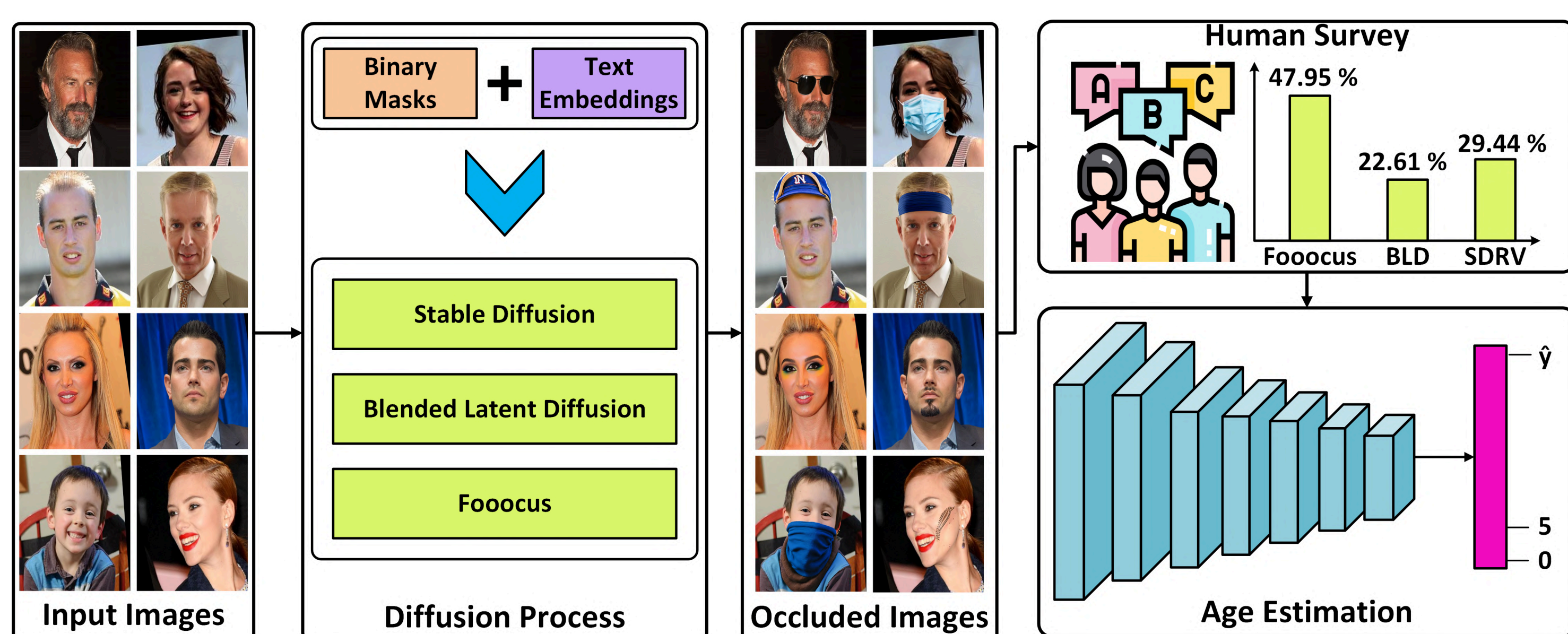
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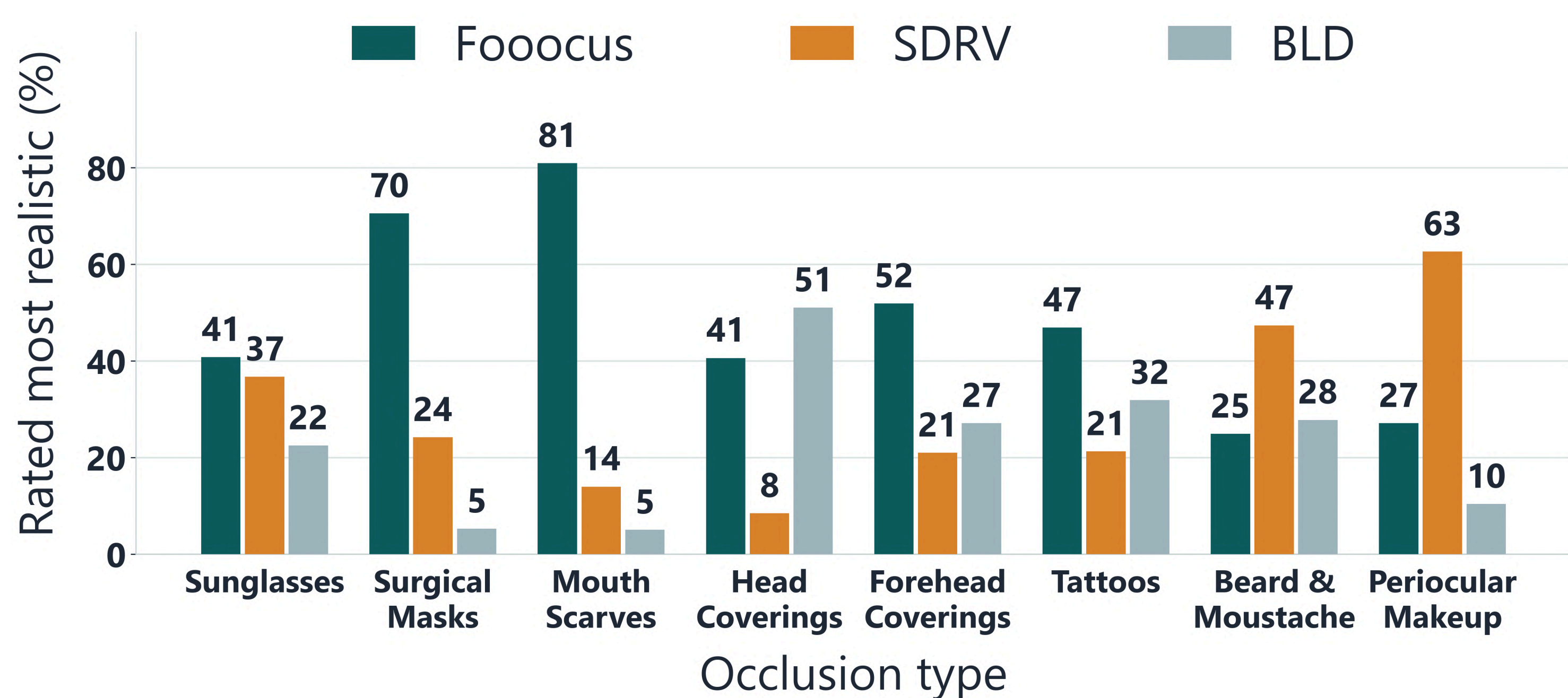
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1 Problem & Approach

- Real-world age estimation struggles with facial occlusions (masks, sunglasses, scarves, etc.).
- No public benchmark dataset exists to evaluate age estimation under occlusions.
- We release the first identity-preserving occluded age estimation datasets.



2 Human Survey



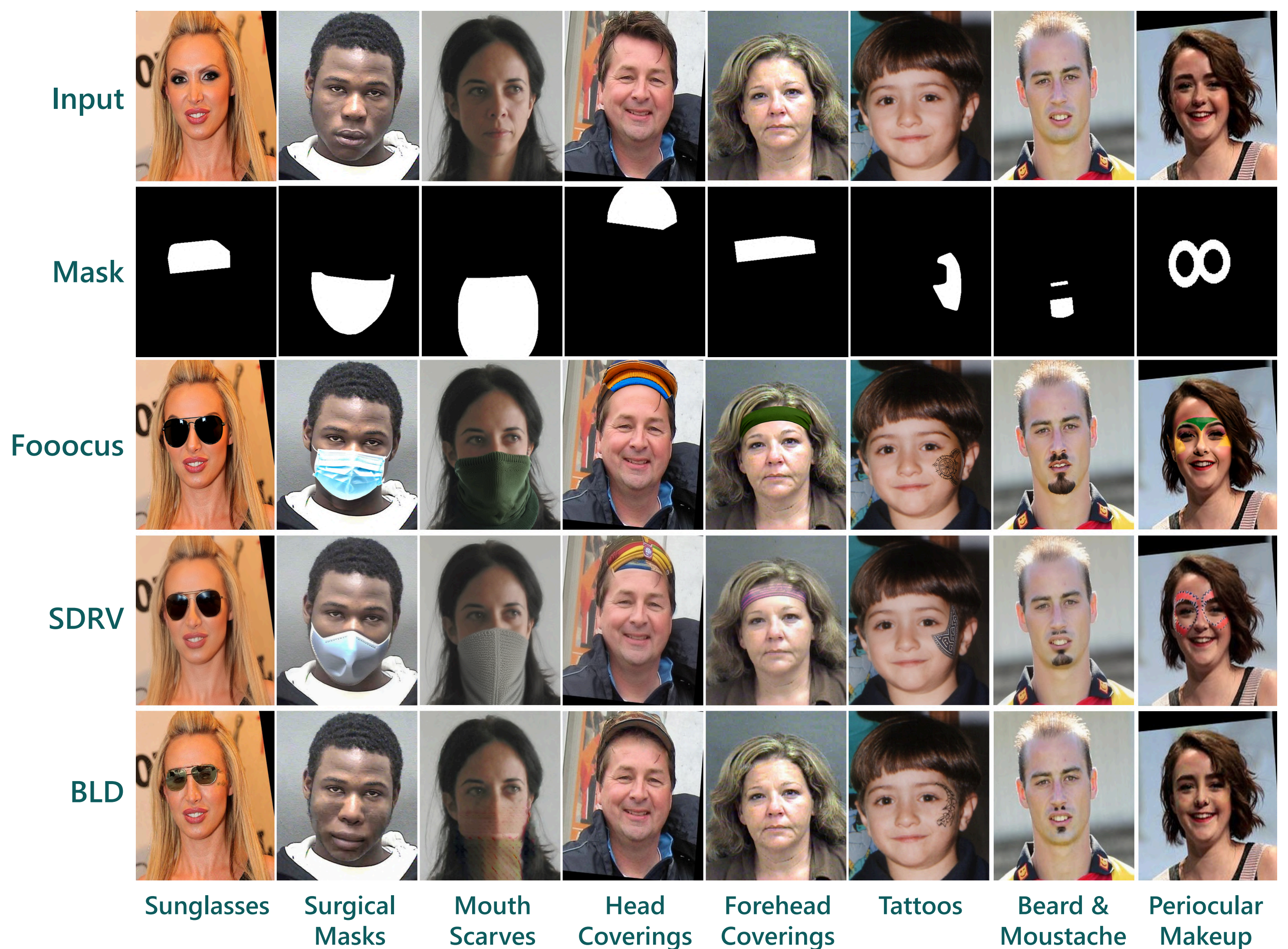
3 Prompts for Occlusion Generation

Occlusion type	Positive keywords	Negative keywords
Sunglasses	aviator, wayfarer, mirrored, polarized	transparent lenses
Surgical Masks	white or light blue medical mask	transparent
Mouth Scarves	neck gaiter, wool muffler, neck warmer	uncovered mouth or neck
Head Coverings	baseball cap, cowboy hat, fedora	deformed head
Forehead Coverings	sweatband, headband, bandana	narrow strip
Tattoos	tribal tattoo, realistic scar, floral	inflamed wound
Beard-Moustache	goatee, handlebar moustache, stubble	unnatural color
Periocular Makeup	bright red, vivid green, bold yellow	altered eyes or iris

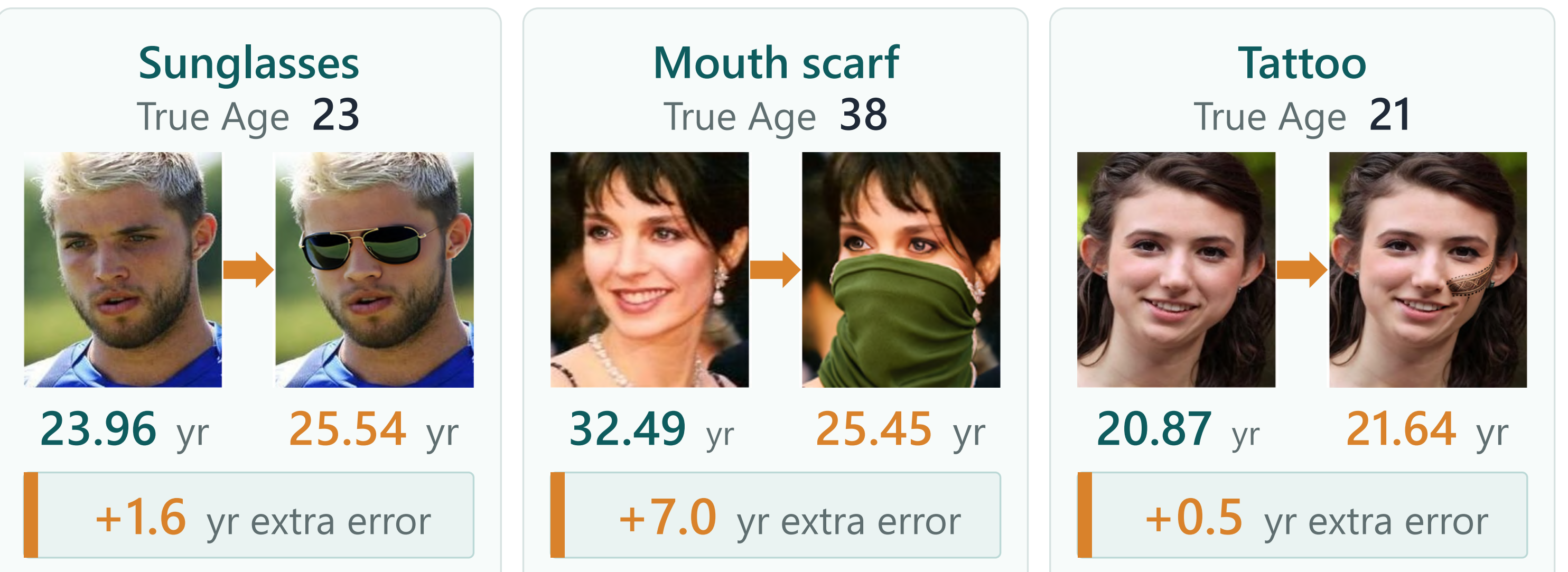
IN EVERY POSITIVE PROMPT
preserving facial identity

IN EVERY NEGATIVE PROMPT
blurry, low quality, distorted face

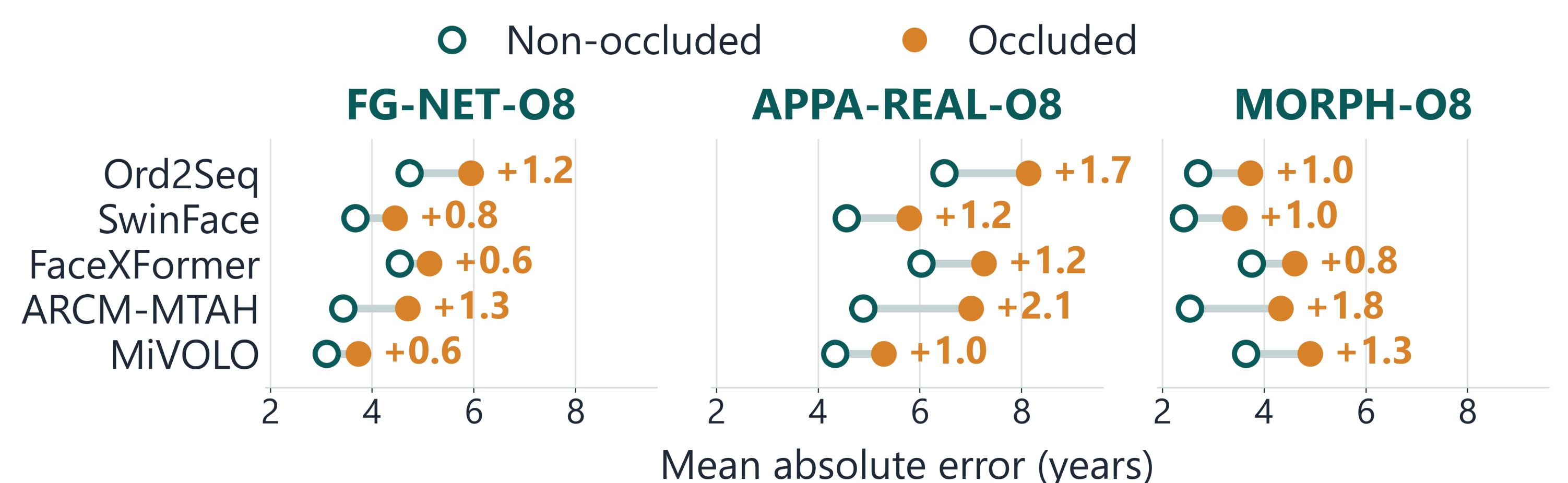
4 Types of Occlusions



5 Results

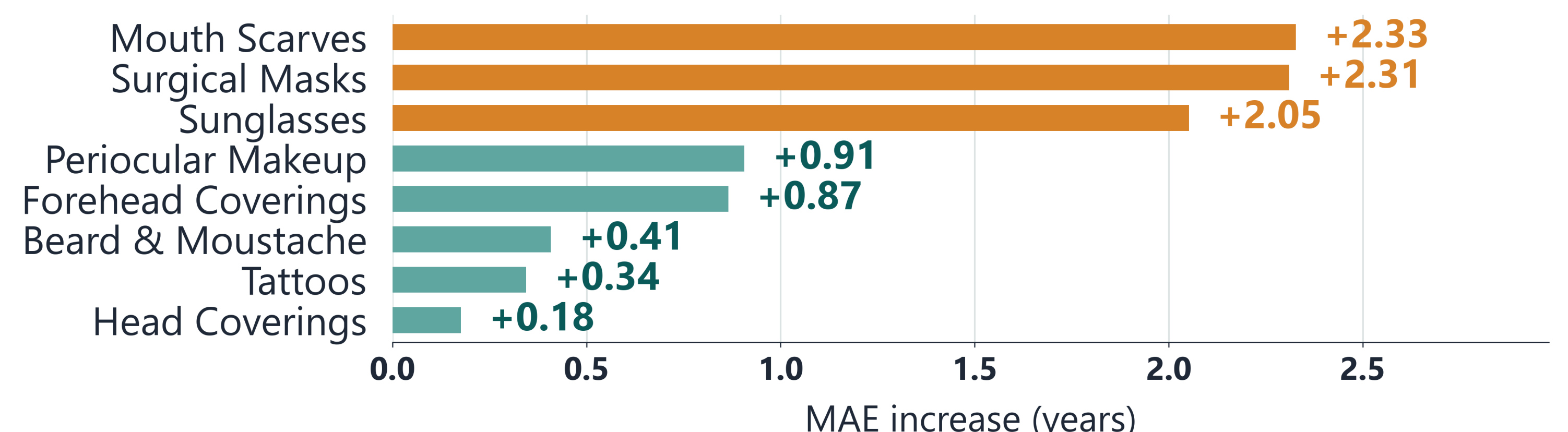


MAE per method and dataset, averaged over the 8 occlusion types



6 Which Occlusions Hurt Most?

MAE increase per occlusion type, averaged over 5 methods × 3 datasets



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